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Exp No: 5	Experiments on Packet capture tool: Wireshark

~~AIM:~~

Tounderstand the features of Wireshark as a packet capture tool and to understand the encapsulation of information at the various layers of a protocol stack.

ALGORITHM / PROCEDURE:

1. Install andopen Wireshark.The home screen lists all the network interfaces with a small activity graph.
2. Double click the active interface (Ethernet / Wi-Fi) to start the capture.
3. Generate some traffic: open a website in the browser, run ping google.com and nslookup example.com in the Command Prompt.
4. Stop the capture using the red square button.
5. Apply display filters in the filter bar, for example icmp, dns, tcp.port == 80, ip.addr == `192.168.1.25.`
6. Select one packet and expand its layers in the **Packet Details** pane. Study the Frame, Ethernet II, Internet Protocol, TCP / ICMP headers.
7. Select any header field and see the corresponding bytes highlighted in the **Packet Bytes** pane.
8. Use **Analyze > Follow > TCP Stream** to follow the full TCP conversation and note the three-way handshake (SYN, SYN-ACK, ACK).
9. Save the capture using**File > Save As (.pcapng)**.

Features of Wireshark

- Captures live packetsfrom Ethernet, Wi-Fi, loopback and other interfaces and also reads saved .pcap files.
- Capture filters (before capture) and display filters (after capture).
- Colouring rules to identify the protocols easily(TCP, UDP, ICMP, ARP, HTTP ...).
- Shows all the protocol headers layer by layerwith the raw bytes.
- Statistics tools like Conversations, ProtocolHierarchy and I/O Graph.

Layer	Protocol example	PDU name	Header added
Application	HTTP, DNS	Data / Message	Application header
Transport	TCP, UDP	Segment / Datagram	Port numbers, sequence no. (TCP 20 B, UDP 8 B)
Network	IP, ICMP	Packet	Source and destination IP address (20 B)
Data link	Ethernet	Frame	Source and destination MAC (14 B) + FCS (4 B)
Physical	UTP, Wi-Fi	Bits	None - only signals

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DIAGRAM:

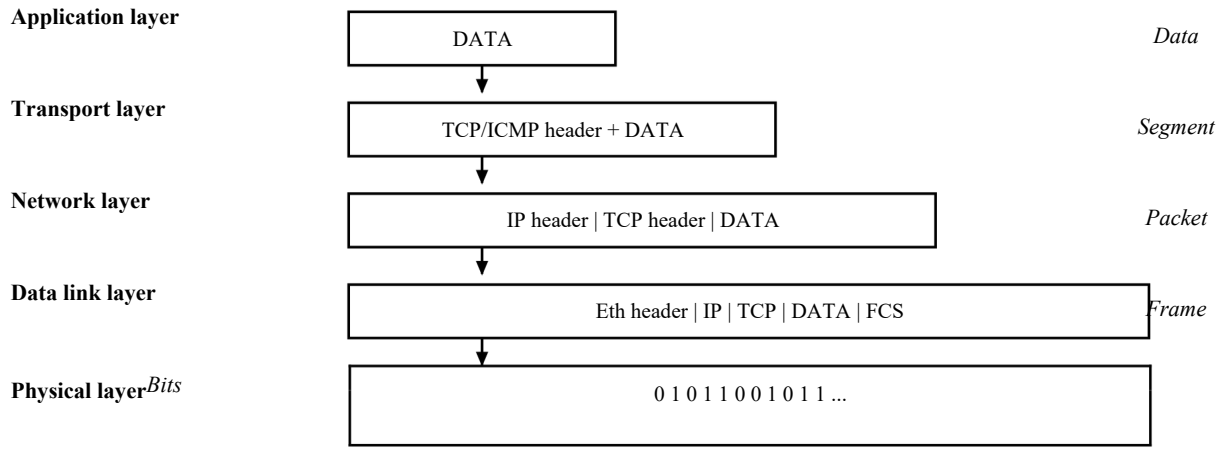


Fig 1: Encapsulation of data as it moves down the protocol stack

OUTPUT:

CapturingfromEthernet - Wireshark

No.	Time	Source	Destination	Protocol	Len	Info
1	0.0 000 00	192 .16 8. 1.2 5	192 .16 8. 1.1	DNS	70	Standard query 0x3a1c A example.com
2	0.0 124 31	192 .16 8. 1.1	192 .16 8. 1.2 5	DNS	86	Standard query response 0x3a1c A example.com A 93.184.21
3	0.0 131 02	192 .16 8. 1.2 5	93. 184 .2 16.	TCP	66	51322 > 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460
4	0.0 418 75	93. 184 .2 16. 34	34 192 .16 8.	TCP	66	80 > 51322 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0
5	0.0 419 32	192 .16 8. 1.2 5	1.2 5 93. 184 .2	TCP	54	51322 > 80 [ACK] Seq=1 Ack=1 Win=131328 Len=0
6	0.0 422 10	192 .16 8. 1.2 5	16. 34 93. 184	HTTP	142	GET / HTTP/1.1
7	0.0 716 55	93. 184 .2 16. 34	.2 16. 34 192	TCP	54	80 > 51322 [ACK] Seq=1 Ack=89 Win=65535 Len=0
8	0.0 933 40	93. 184 .2 16. 34	.16 8. 1.2 5 192	HTTP	1294	HTTP/1.1 200 OK (text/html)
9	1.402118	192 .16 8. 1.2 5	.16 8. 1.2 5 142	ICMP	74	Echo (ping) request id=0x0001, seq=21/5376, ttl=128
10	1.4 168 75	142 .25 0. 195 .7 8	.25 0. 195 .7 8	ICMP	74	Echo (ping) reply id=0x0001, seq=21/5376, ttl=117
11	2.118004	c4:e9:84:2a:5b:10	192 .16 8. 1.2 5	ARP	60	Who has192.168.1.25?Tell 192.168.1.1

Packet List pane - captured packets (colours show the protocol)

Frame 9: 74 bytes on wire (592 bits), 74 bytes captured

Packet Details - Frame 9 (ICMP Echo request)

```
v Frame 9: 74 bytes on wire (592 bits), 74 bytes captured (592 bits) on interface
\Device\NPF_{8F3A21B0}   , id 0
    Arrival Time: Oct 7, 2026 20:57:38.402118000 India Standard Time
    Frame Length: 74 bytes (592 bits)
Protocols in frame: eth:ethertype:ip:icmp:data
v Ethernet II, Src: 00:1a:2b:3c:4d:5e, Dst: c4:e9:84:2a:5b:10
Destination: c4:e9:84:2a:5b:10
    Source: 00:1a:2b:3c:4d:5e     Type: IPv4 (0x0800) v Internet
Protocol Version 4, Src: 192.168.1.25, Dst: 142.250.195.78     0100
.... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Total Length: 60
    Identification: 0x7a31 (31281)
    Time to Live: 128
    Protocol: ICMP (1)
    Source Address: 192.168.1.25
Destination Address: 142.250.195.78 v
Internet Control Message Protocol
Type: 8 (Echo (ping) request)
    Code: 0
    Checksum: 0x4d5a [correct]
    Identifier (BE): 1 (0x0001)
    Sequence Number (BE): 21 (0x0015)
    Data (32 bytes)
```

Packet Bytes - Frame9

```
0000  c4 e9 84 2a 5b 10 00 1a 2b 3c4d5e08004500 00 3c  ...*[...+<M^..E.
0010  7a 31 00 00 80 01 00 00c0a801198efa c3 4e 08 00  .<z1............
0020  4d 5a 00 01 00 15616263646566 67 68 69 6a 6b 6c  .N..MZ....abcdef
0030  6d 6e 6f 70717273747576 77 61 62 63 64 65 66 67  ghijklmnopqrstuv
0040  68 69                                           wabcdefghi
```

Layer Ethernet II	Bytes in the frame	Observed value
	00 - 13 (14 bytes)	`c4:e9:84:2a:5b:10 <- 00:1a:2b:3c:4d:5e, type 0x0800`
IPv4	14 - 33 (20 bytes)	`192.168.1.25 -> 142.250.195.78, TTL 128, protocol 1 (ICMP)`
ICMP	34 - 41 (8 bytes)	`Type 8, Code 0, id 1, seq 21 abcdefghijklmnopqrstuvwabcdefghi`
Data	42 - 73 (32 bytes)	

RESULT:

Thus thefeatures of Wireshark were studied. Packets were captured and the encapsulation of data at the Data link (Ethernet), Network (IP), Transport (TCP) and ICMP layers was observed in the packet details.